

HTML Structure & CSS Layout Foundations

Semantic Document Engineering and Data Table Presentation

Institute Handout — Systems & Software Architecture Group

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Lab Objectives

By the end of this intensive 2-hour lab session, students will be able to:

1. Build a syntactically correct HTML5 semantic document boilerplate from scratch.
2. Implement structural and text hierarchy using headings, blocks, paragraphs, and list arrays.
3. Construct functional data matrices using tabular tags (`table`, `tr`, `th`, `td`).
4. Intercept and manipulate visual layouts using inline styles and embedded internal CSS selectors.
5. Map elements using layout mechanics, custom border properties, and structural cell padding.

Safety Warning for Beginners

Case Sensitivity & Nesting Strictness:

- **Improper Nesting:** Elements must close in the reverse order they were opened (e.g., `<b><i>text</i></b>`). Overlapping tags break document parsing models and distort page layouts.
- **File Extensions:** Ensure you explicitly save files with a clean `.html` suffix. Saving a document as `index.html.txt` will render raw code in the browser instead of the interpreted document structure.

1 Task 1: The Core HTML5 Boilerplate Skeleton (25 Mins)

Before rendering visual details or tables, you must declare an explicit structural document template so internet browsers understand how to process your components.

1. Open your terminal application, move onto your desktop environment, and instantiate a fresh project directory layout:

```
cd Desktop
mkdir html_core_lab
cd html_core_lab
```

2. Create an empty entry-point document file using your system's native console command strings:

Command Prompt (CMD):

```
type nul > index.html
```

PowerShell (PS):

```
New-Item index.html
```

3. Open `index.html` in your preferred plain-text or code editor. Assemble the fundamental document structure by writing out the exact template skeleton shown below:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-
scale=1.0">
  <title>Developer Sandbox Profile</title>
</head>
<body>

  <!-- Content implementations will reside inside here -->

</body>
</html>
```

4. Double-click the `index.html` file from your system file explorer window to launch it within a web browser. Verify that your application tab displays "Developer Sandbox Profile" on its header line.

2 Task 2: Semantic Elements & Content Layout Blocks (25 Mins)

Let's populate the body container with core typographical elements. This structural hierarchy will serve as the foundation when we connect data streams down the line.

1. Add a high-level site title and a brief introductory paragraph within your `<body>` container tags:

```
<h1>System Developer Profile Portfolio</h1>
<p>Welcome to the isolated frontend engineering environment sandbox
context.</p>
```

2. Implement an unordered list element structure to outline your core primary technical disciplines:

```
<h2>Primary Engineering Competencies</h2>
<ul>
  <li>Python Runtime Backend Engineering</li>
  <li>Relational Database Architecture Design</li>
  <li>Structured Semantic Layout Development</li>
</ul>
```

3. Refresh your open web browser window. Observe the default system typography layouts, element separation behavior, and natural spacing offsets applied natively by the rendering engine.

3 Task 3: Tabular Grid Matrices for Structured Datasets (35 Mins)

Web components often require a way to present complex datasets cleanly. We will build a structured data grid array using HTML's native tabular tags.

1. Directly beneath your list component block, add a secondary structural section header:

```
<h2>Active Systems Architecture Inventory Matrix</h2>
```

2. Initialize a complete multi-column tracking grid array. Use header definitions (<th>) to name each column and data cells (<td>) to specify row properties:

```
<table>
  <thead>
    <tr>
      <th>Application Layer ID</th>
      <th>Technology Framework Stack</th>
      <th>Infrastructure Role Variant</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>APP-001</td>
      <td>Django Web Framework</td>
      <td>Main Core Portal Service Backend</td>
    </tr>
    <tr>
      <td>APP-002</td>
      <td>PostgreSQL Engine</td>
      <td>Relational Transaction Persistence Storage</td>
    </tr>
  </tbody>
</table>
```

3. Save your progress and refresh your active browser view. Note that while the data aligns neatly into rows and columns, it lacks borders, padding, and visual definition.

4 Task 4: Manipulating Document Presentation using CSS (35 Mins)

Now we will use Cascading Style Sheets (CSS) to add explicit styling to our elements, modify structural spacing, and clarify our tabular layouts.

4.1 Part A: Local Inline Element Override Attributes

Inline CSS overrides specific elements directly within the body container, making it ideal for targeted visual adjustments.

1. Apply a direct structural inline color value modification property onto your main top heading element:

```
<h1 style="color: #1F4E79;">System Developer Profile Portfolio</h1>
```

4.2 Part B: System Document Styling via Internal CSS blocks

To style repeating elements across the page consistently without writing repetitive inline attributes, we use an internal style block within the document head context.

1. Locate your `<head>` container region at the top of `index.html`. Append an explicit structural `<style>` section block directly below the title line element:

```
<head>
  <meta charset="UTF-8">
  <title>Developer Sandbox Profile</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 40px;
      background-color: #F5F7FA;
    }
    table {
      width: 100%;
      border-collapse: collapse;
      margin-top: 20px;
    }
    th, td {
      border: 1px solid #D2D7DF;
      padding: 12px;
      text-align: left;
    }
    th {
      background-color: #1F4E79;
      color: white;
    }
  </style>
</head>
```

2. Save your changes and refresh your browser. Analyze the visual enhancements: the fonts are cleaner, the workspace margins are evenly spaced, and the data grid features borders and high-contrast table headers.

5 Lab Exit & Comprehensive Student Cleanup Sequence

To conclude your practical development session, execute these steps to safely clear your testing workspace and wipe down development states.

1. Close the web browser application window that is actively rendering your sandbox profile file.
2. Shut down your active text editor interface tools and ensure any pending file hooks are detached.
3. Return to your command line terminal. Step back up and completely remove the sandbox folder tree:

```
cd ..
```

4. Permanently purge your test directories and files. *Remember, terminal deletion actions bypass your system Recycle Bin entirely.*

Command Prompt (CMD):

```
rmdir /s /q html_core_lab
```

PowerShell (PS):

```
Remove-Item -Recurse -Force html_core_lab
```

5. Wipe down your shell window screen history completely before presenting your workstation setup state to your tracking lab supervisor for final session grading marks:

Command Prompt (CMD):

```
cls
```

PowerShell (PS):

```
Clear-Host
```